Shaan Pakala

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About

I am a 2nd year Ph.D. student in the Computer Science department at University of California, Riverside. I am advised by Prof. Vagelis Papalexakis, and we use tensor methods to address problems in AI4Science research. Some key examples are (i) interpretable surrogate modeling for designing materials and (ii) efficiently generating physics simulations by outputting their low-rank tensor factors with diffusion models and GANs. In my research, I regularly collaborate with domain-expert researchers at Lawrence Livermore National Laboratory (LLNL), the LIGO Scientific Collaboration, as well as various healthcare institutions.

Education

University of California, Riverside

- **Ph.D. in Computer Science**, advised by Prof. Vagelis Papalexakis — **Sept. 2025 – Present**
- **B.S. in Data Science** — **Sept. 2021 – June 2025**

Research Experience

Nokia Research

- **AI R&D Intern** — **Fall 2026**
  Intern in the applied AI research group, with Dr. Liang Wu. Working on implementing safe & robust agentic AI for internal company use.

Lawrence Livermore National Laboratory

- **Research Intern** — **Summer 2026**
  Intern in the predictive healthcare group, with Dr. Priyadip Ray. I worked on using machine learning for early disease outbreak detection, using the Tuolumne supercomputer.
- **Research Intern** — **Summer 2025**
  Intern in the predictive healthcare group, with Dr. Priyadip Ray and Dr. Braden Soper. I worked on causal inference for Parkinson's Disease progression, with clinicians/biomedical researchers.

University of California, Riverside

- **Graduate Student Researcher** — **June 2025 – Present**
  I work with domain-experts in physics & material science on AI4Science problems using tensor decomposition: (i) surrogate modeling for robust & interpretable material design [1] [3] and (ii) efficient physics simulation generation using tensor decomposition in GANs & diffusion models [4].
- **Undergraduate Student Researcher** — **June 2024 – June 2025**
  Research with Prof. Papalexakis. I worked on automating the optimization of data science pipelines (e.g., neural network architecture search, SQL queries) with tensor completion [2].

Peer-reviewed Papers *(* denotes equal contribution)*

Conference

[1] Tensor Methods: A Unified and Interpretable Approach for Material Design
Shaan Pakala, A.E. Gongora, B. Giera, and E.E. Papalexakis
Proceedings of ACM SIGKDD (AI for Sciences Track) 2026. [PDF] [Code]

[2] Automating Data Science Pipelines with Tensor Completion
Shaan Pakala, B. Graw, D. Ahn, T. Dinh, M.T. Mahin, V. Tsotras, J. Chen, and E.E. Papalexakis
Proceedings of IEEE International Conference on Big Data 2024. [PDF] [Code]

Workshop

[3] Surrogate Modeling for the Design of Optimal Lattice Structures using Tensor Completion
Shaan Pakala, A.E. Gongora, B. Giera, and E.E. Papalexakis
NeurIPS AI for Accelerated Materials Design Workshop 2025. [PDF] [Code]

[4] Efficiently Generating Multidimensional Calorimeter Data with Tensor Decomposition Parameterization
P. Goulart*, **Shaan Pakala***, and E.E. Papalexakis
ICCV Representation Learning with Very Limited Resources Workshop 2025. [PDF] [Code]

[5] Tensor Completion for Surrogate Modeling of Material Property Prediction
Shaan Pakala, D. Ahn, and E.E. Papalexakis
AAAI Bridge on Knowledge-Guided Machine Learning 2025. [PDF]

Awards

- **Travel Award** (KDD'26, SDM'26 Ph.D. Forum, NeurIPS'25 AI4Mat Workshop, BigData'24)
- **Undergraduate Research Spotlight Award** (UC Riverside)
- **NSF Research Experience for Undergraduates** (UC Riverside)

Contributed Talks & Posters

Talks

- **Tensor Methods for Interpretable Material Design** — *LLNL CASIS Workshop*, 2026
- **Surrogate Modeling with Tensor Completion** — *UTRGV MECIS Spring Webinar*, 2026

Posters

- **Exploring Scientific Design Spaces with Tensor Decomposition** — *SIAM SDM Ph.D. Forum*, 2026
- **Tensor Methods for Interferometer Simulations** — *UCR Detector Science Workshop*, 2026
- **Tensor Methods for Interpretable Material Design** — *SAIR UCR Science & AI Summit*, 2026

Academic Service

- **Organizer:** AI4AutoSci Workshop at IEEE BigData'26
- **Reviewer:** Journal of Machine Learning Research (JMLR), TensorKDD Workshop at KDD'26, AI Data Scientist Workshop at KDD'26